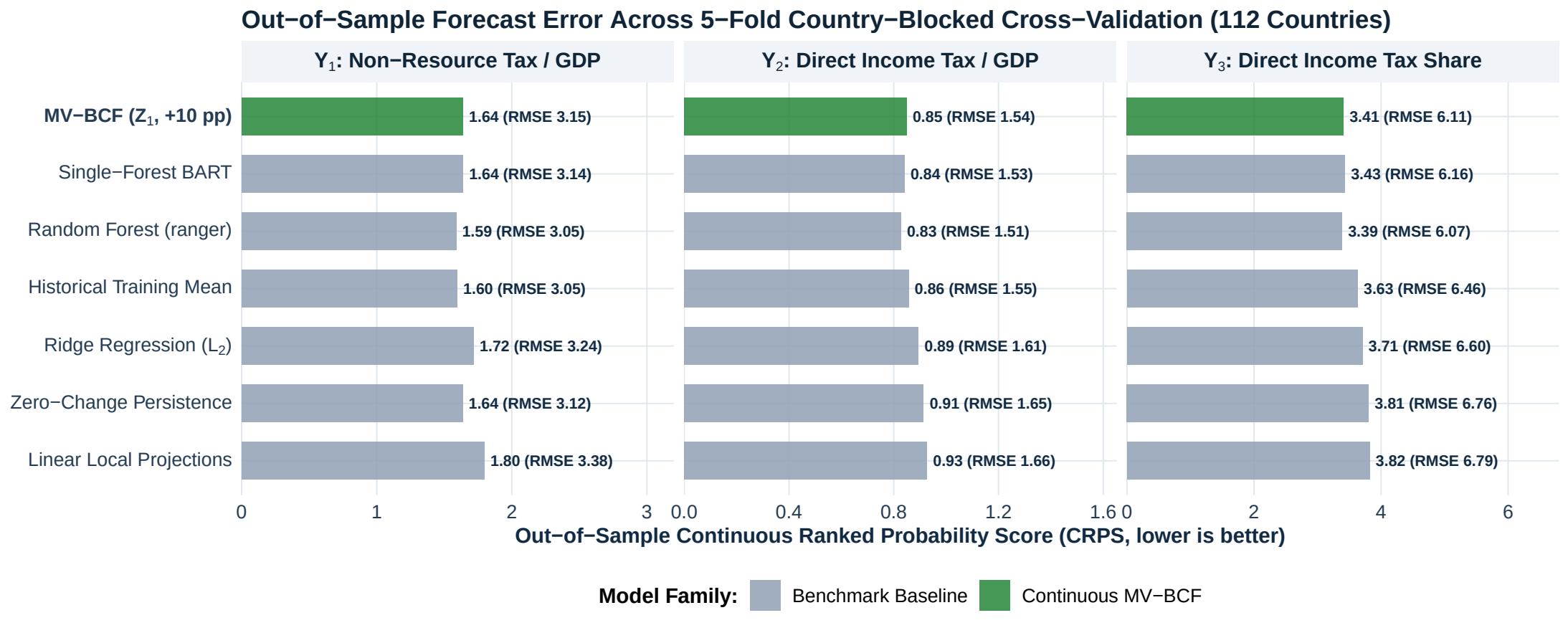
Out−of−Sample Forecast Error Across 5−Fold Country−Blocked Cross−Validation (112 Countries)
Y1: Non−Resource Tax / GDP
Y2: Direct Income Tax / GDP
Y3: Direct Income Tax Share
MV−BCF (Z1, +10 pp)
Single−Forest BART
Random Forest (ranger)
Historical Training Mean
Ridge Regression (L2)
Zero−Change Persistence
Linear Local Projections
1.64 (RMSE 3.15)
1.64 (RMSE 3.14)
1.59 (RMSE 3.05)
1.60 (RMSE 3.05)
1.72 (RMSE 3.24)
1.64 (RMSE 3.12)
1.80 (RMSE 3.38)
0.85 (RMSE 1.54)
0.84 (RMSE 1.53)
0.83 (RMSE 1.51)
0.86 (RMSE 1.55)
0.89 (RMSE 1.61)
0.91 (RMSE 1.65)
0.93 (RMSE 1.66)
3.41 (RMSE 6.11)
3.43 (RMSE 6.16)
3.39 (RMSE 6.07)
3.63 (RMSE 6.46)
3.71 (RMSE 6.60)
3.81 (RMSE 6.76)
3.82 (RMSE 6.79)
0
1
2
3
0.0
0.4
0.8
1.2
1.6
0
2
4
6
Out−of−Sample Continuous Ranked Probability Score (CRPS, lower is better)
Model Family:
Benchmark Baseline
Continuous MV−BCF